

Impact of virtual environments on ability to localize

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The aim of the present study was to validate the use of a virtual system in sound localization tasks. Experiment 1 evaluated the impact of virtual sounds (*generic vs individualized*) in sound localization. To this end, the Minimum Audible Angle was collected for different azimuths and virtual sound data bases. The results indicate that we do not need to use individualized virtual sound. Experiment 2 evaluated the impact of virtual sound and visual environments in sound localization task. The results validated the use of generic virtual sounds in sound localization tasks. However, that experiment did not establish the impact of visual environment in subject's performances. Experiments validated the use of this virtual system in sound localization tasks. In the future, this system could be used in order to train hearing impaired patients in sound localization tasks.

INTRODUCTION

In humans, the ability to locate the source of sound contributes to communication. How humans identify the source of sounds is determined by three acoustic cues: (a) Interaural Time Difference (ITD), (b) Interaural Level Difference (ILD) and spectral cues derived from the shape of one's ears, head and torso (Møller et al., 1995; Majdak et al., 2014). Together, these cues provide us with a fairly accurate representation of acoustic space (Sabin et al., 2005). In order to simulate natural acoustic perception, these acoustic cues are usually produced using Head Related Transfer Functions (HRTFs), which can either be generic (e.g., from an acoustic mannikin) or individualized (considering morphology). The use of HRTFs leads to accurate source localization and an increased sense of presence within a virtual environment, when compared to non-spatialized audio (Hendrix and Barfield, 1996; Bergstrom, 2017).

Each of us has specific morphological characteristics, thus each of us has specific HRTFs. However, creating individualized HRTFs can take a long time, be technically difficult and expensive to implement (Meshram et al., 2014). In experiment 1, we evaluated the impact of virtual sounds (*generic versus individualized*) in sound localization. To this end, we collected the Minimum Audible Angle (MAA), which is a measurement of the resolution of auditory space. We assumed that individualized HRTFs lead to a better resolution of auditory space. Research on multisensory integration (Witten and Knudsen, 2005; Ghazanfar and Schroeder, 2006; Stein and Stanford, 2008) has highlighted the extent to which visual perception can influence auditory perception (Howard and Templeton, 1966; Vroomen et al., 2001; Bonath et al., 2007). One classic example of influence of the visual over the perceived

location is the ventriloquist illusion - an audio-visual illusion in which the perceived location of an auditory source is translocated toward a visual source that is presented at the same time, but in a different location (Howard and Templeton, 1966; Bertelson and Aschersleben, 1998). In experiment 2, we explored the impact of virtual environment in sound localization tasks. To this end, we simulated different environments, with more or fewer visual references. During this experiment, sound stimuli were virtual and chosen according to experiment 1. We presumed that adding visual reference would lead to an improvement in sound localization tasks.

EXPERIMENT 1

Methods

Subjects. 14 subjects took part in the experiments (4 males and 10 females; age range, 21-29). They had normal audiometric thresholds equal to or below 20 dB hearing loss at the octave-band frequencies between 250 Hz and 8 kHz. They had not previously performed localization tests.

Material. HRTFs provided by 4 different data bases.

Generic data bases

Among the data bases, 2 of them were generic. This means that data bases are extracted from acoustic mannikins. One came from the **KEMAR** (Knowles Electronics Mannikin for Acoustic Research) (Gardener & Martin, 1995), the other one from **KU100** (Bomhardt et al., 2016).

Individualized data bases

2 individualised data bases determined with Bomhardt's data (Bomhardt et al., 2016). This study collected HRTFs from 43 subjects, associated with anatomic measurements for each subject.

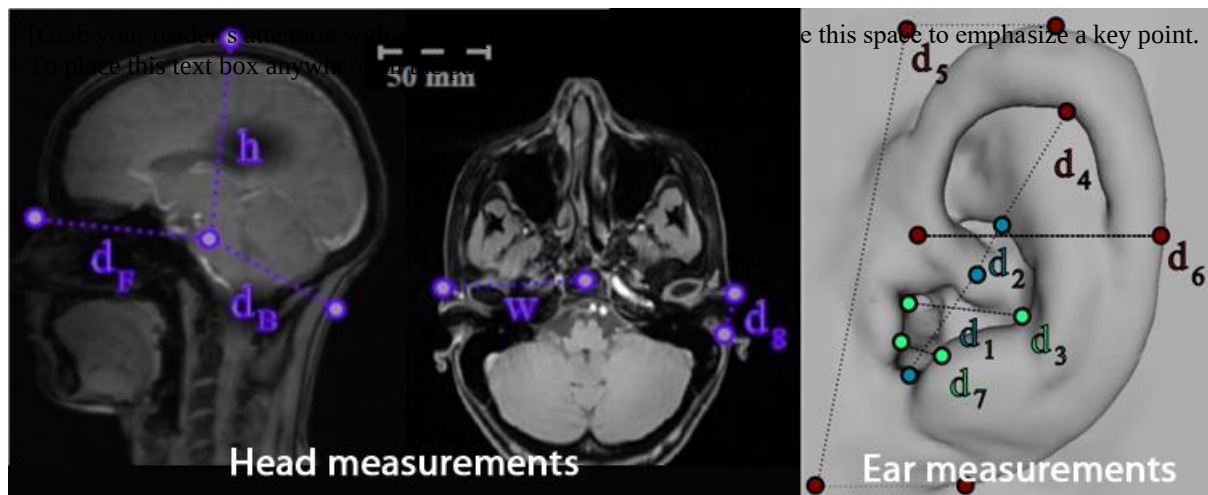


Figure 1 Anthropometric measurements for the ITA and ITA1 data bases. 4 head measurements (h , d_E , d_B , and W) and 7 auricle measurements (d_1 , d_2 , d_3 , d_4 , d_5 , d_6 and d_7) were used to estimate the subjects' head (reproduced from Bomhardt, 2016).

In order to individualize HRTF data, we collected measurements from our subjects (Figure 1). We created 2 vectors (**ITA** and **ITA1**) from the 43 Bomhardt subjects and our 14 subjects. ITA gathered head measurements and ear measurements and ITA1 only ear measurements.

MAAs were tested for those 4 data bases binaurally. In addition, we evaluated MAA in monaurally conditions: **Ipsilateral Attenuation** (i.e. on the same side as the deviant sound) and **Contralateral Attenuation** (i.e., on the opposite side to the deviant sounds).

Experimental procedure. The MAA task was conducted using a two-alternative forced choice design with a 1-up/1-down procedure. During the test, participants were required to differentiate between one standard signal (signal from the same position, i.e. 0° , -50° or 180°) and one deviant signal that differed in its position. The standard locations for the signals were at 0° , -45° and -180° ("-" indicates the left side).

During the test, participants were required to differentiate between two standard signals (signals from the same position) and one deviant signal that differed in its position.

Statistical analyses. A repeated-measures analysis of variance (RM-ANOVA) was chosen for group comparisons as the basis for considering the potential effects of other co-variables. In the present study, the RM-ANOVA test was performed to examine the effects of data banks and azimuth on MAA thresholds.

The mean value and standard deviation of MAA thresholds were calculated and compared using a post-hoc t-test when significant differences were found.

Statistical analyses were performed using the R studio software (R Development Core Team (2005)). A $p < 0.05$ was considered to be statistically significant.

Results

The results of the experiment are shown in Figure 2. Values of MAA performance were subjected to a 3 (azimuth: 0° , -50° , -180°) $\times$ 6 (data banks: KEMAR, KU100, ITA, ITA1, Att. Contra, Att. Ipsi) repeated-measures ANOVA. There were significant main effects of azimuth [$F(1,54 ; 20,07) = 11,61$, $p = 0,0009$] and data banks [$F(3,27 ; 42,52) = 26,68$, $p = < 0,0001$]. Significant interaction was found between the factors of azimuth and data banks [$F(5,89 ; 76,53) = 2,70$, $p = 0,02$]. A post-hoc test was performed.

Effect of Azimuth. The post-hoc test on variable azimuth showed no significant differences between position 0° and position 180° [$t(26) = -0,859$, $p = 0,6701$]. Comparison between 0° - 50° [$t(26) = 3,677$, $p = 0,0030$] and 180° - 50° [$t(26) = 4,536$, $p = 0,0003$] showed significant differences. Comparisons for each pair were performed.

- They confirmed the absence of significant differences between position 0° and position 180° in binaural conditions, regardless of data banks,
- MAA was lower at 0° and 180° compared to 50° , regardless of data banks or attenuation (monaural and binaural).

Effect of monaural attenuation. MAAs were significantly lower in binaural conditions compared to monaural attenuation.

- Ipsilateral and contralateral conditions (i.e., monaural conditions) did not show any significant differences.

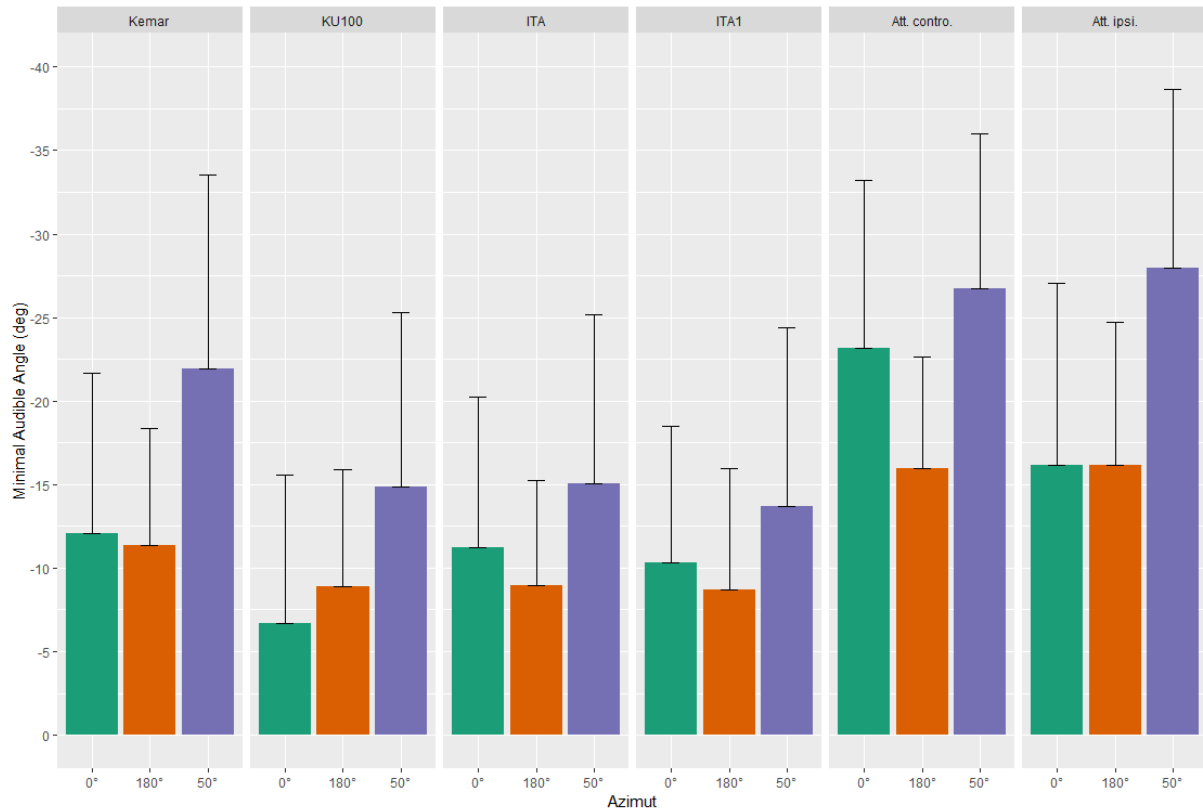


Figure 2 Minimum Audible Angle for the different data banks at various different azimuths. 4 data banks were tested binaurally (Kemar, KU100, ITA and ITA1), while 2 were tested monaurally (Att. Contro. And Att. Ipsi.).

Effect of data bases. Overall, MAA for different data banks did not differ.

- Only MAAs for Kemar data banks were significantly higher compared to the others.
- Other comparisons between data banks did not show significant differences.

Discussion

MAAs collected during the experiments showed:

- An absence of significance differences between data banks,
- An absence of significance differences for MAA means between data banks,
- Significance differences between positions 0°-180° and position 50°,
- Reduction of MAAs with monaural attenuation,
- Side attenuation did not affect MAAs amplitude.

Those results were compared to existing studies.

MAA amplitude. The results in the present study were as follows:

- **-6.69** degrees (KU100) to **-12.04** degrees (KEMAR) for 0° position,
- **-8.69** degrees (ITA1) to **-11.36** degrees (KEMAR) for 180° position,
- **-13.71** degrees (ITA1) to **-21.94** degrees (KEMAR) for 50° position.

The trends we observed (MAAs similar for 0° and 180°, MAA lower at 0° and 180° compared to 50°) were similar to existing studies (Kidd et al., 2005; Middlebrooks & Green, 1991). However, MAAs in existing studies are lower than ours (MAAs range between 2.5° (0°), 6° (45°) and 5° (90°) for normal-hearing subjects. This may originate from our protocol. Cai et al. (2015) collected MAAs means in free field with speakers. The use of earphones could degrade some monaural features which are essential for localization at 0° and 180°. Resonance phenomenon, induced by the use of earphones, could also impact MAA amplitude.

Simulation of unilateral hearing loss.

Implementation of attenuation was used to simulate unilateral hearing loss. The MAAs collected were compiled as follows:

- **-23.15** degrees (contralateral attenuation) and **-16.13** degrees (ipsilateral attenuation) for 0° position,
- **-15.97** degrees (contralateral attenuation) and **-16.13** degrees (ipsilateral attenuation) for 180° position,
- **-26.74** degrees (contralateral attenuation) and **-27.96** degrees (ipsilateral attenuation) for 50° position.

Cai et al. (2005) performed MAA measurements for unilateral hearing loss in free field:

- **-13** degrees (left hearing loss) and **-8** degrees (right hearing loss) for 0 degrees,
- **-17.5** degrees (left hearing loss) and **-16** (right hearing loss) for 45 degrees.

Previous studies found lower MMAs for unilateral hearing-impaired subjects compared to normal-hearing subjects.

Use of Database. The present study was not arguing for differences in MAA amplitude between the different data banks (generic or individualized). To our knowledge, there are no publications which compare MAAs between different data banks.

Conclusion

In this experiment we collected MAAs with a new protocol for different data banks at different azimuths.

The starting hypothesis was that individualized HRTFs lead to a better resolution of auditory space. The results did not confirm is hypothesis. According to our results, the use of individualized HRTFs is not necessary.

The Minimum Audible Angle (MAA) measures the resolution of auditory space. That measurement might have a clinical use. Indeed, implementation of that measurement is low cost because it requires only a computer and ear-phones.

EXPERIMENT 2

Methods

Subjects. 6 subjects took part in the study (3 males and 3 females; age range, 20-23). They had normal audiometric thresholds equal to or below 20 dB hearing loss at the octave-band frequencies between 250 Hz and 8 kHz. They had not previously performed localization tests.

Material. Sound stimuli were used depending on the results of Experiment 1. We used the KU100 data base. We explored sound localization ability with 7 positions: central position (0°), and 3 positions on both sides (-90°, -60°, -30° (left side) – 30°, 60°, 90°

(right side)). The auditory stimuli lasted 50 msec sent, through Pioneer SE-CH5T earphones. Visual environments were simulated using Unity software (2019.1). During the experiment, the subjects wore a HTC Vive helmet for visual stimuli.

3 different environments were simulated, with more or fewer visual references: i) without visual reference (**Monochrome**) ii) with few visual reference (**Grid**) and iii) a lot of visual references (**Speakers**, with 13 targets corresponding with azimuth -90°, -75°, -60°, -45°, -30°, -15°, 0°, 15°, 30°, 45°, 60°, 75° and 90°) (Figure 4).

Procedure. Prior to the actual test, subjects got to know the virtual sound and visual environments. The test was divided into 3 blocks. Each block represented a visual environment, and the order of passage was randomized. Each block included 35 trials.

Each trial unfolded as follows:

- First, the virtual sound was presented. During the auditory stimulation, the subject stood in front of the azimuth 0°,
- After the auditory stimulation, the subject could move their head. With the remote control, the subject could indicate the target,
- Finally, he came back to initial position.

Data analysis. Firstly, data verification was performed. The main goal of the verification was to harmonized the 0 reference for each participant. Cartesian coordinates were converted into polar coordinates, and the angle was extracted for each trial. The front position (0°) was taken as reference: a “-” was added for a left-side position, a “+” was added for a right-side position.

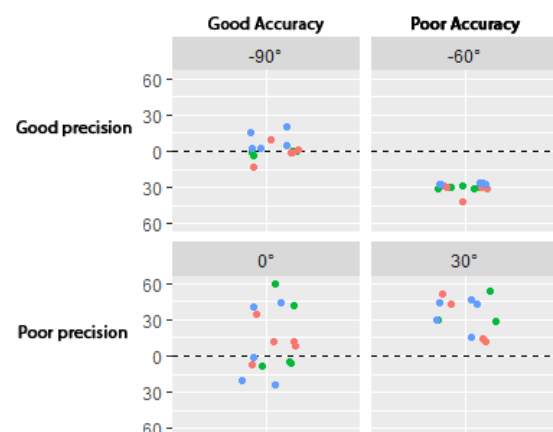


Figure 3 illustrates 2 important concepts in sound localization research. Four examples show subjects' answers, and illustrate the 4 different combinations of accuracy (columns) and precision (rows) that might result.

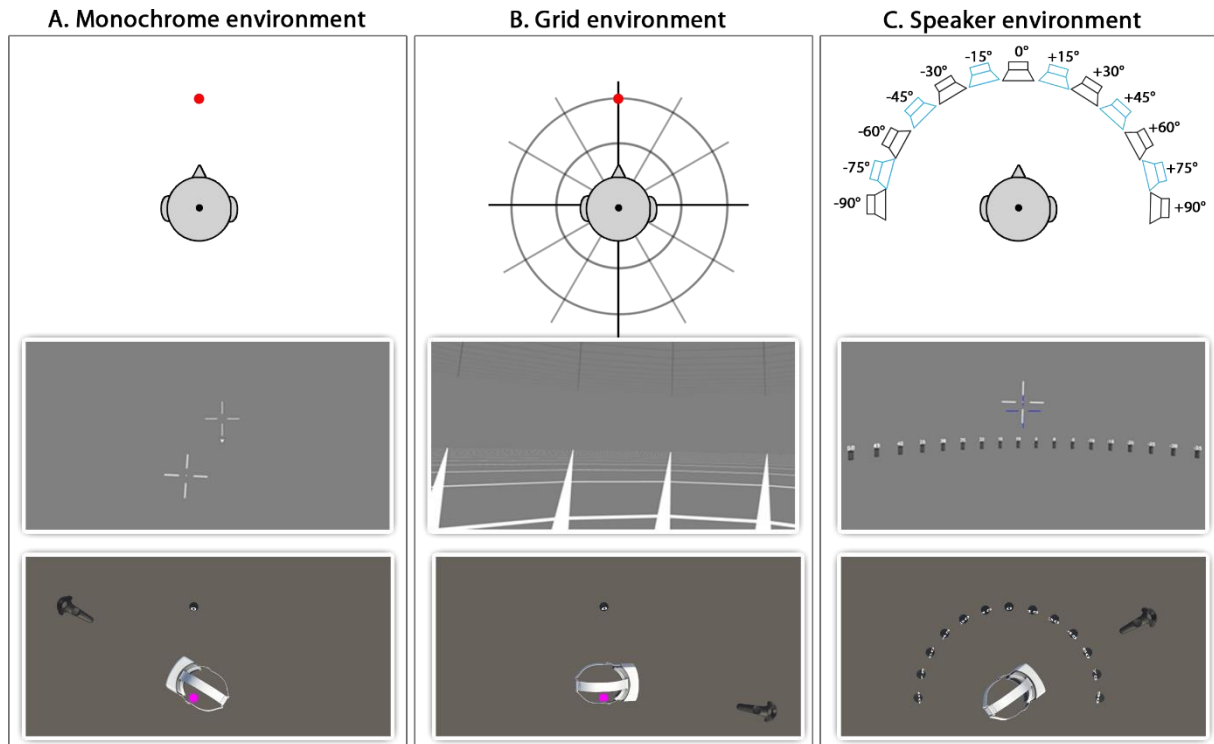


Figure 4 Different environments for the experiment (A. Monochrome environment – without visual reference, B. Grid environment – few visual references, C. Speaker environment – many visual references). From top to bottom, are represented i) theoretical point of view, ii) subject point of view and iii) experimenter point of view.

The errors were calculated by subtracting the target angles from the response angles. Following the conventions of Heffner and Heffner (2005), we distinguished between the *lateral bias* - the systematic error - and *lateral precision error* —the statistical error. The *lateral bias*, sometimes called the localization accuracy, is the mean signed mean of the lateral errors. The *lateral precision error* is the standard deviation of the lateral errors. It represents the consistency of localization ability and is sometimes called localization blur (Figure 3).

Statistical analysis. 2 repeated-measures analysis of variance (RM-ANOVA) were chosen for group comparisons as the basis for considering the potential effects of other co-variables. In the present study, the RM-ANOVA test was performed to examine the effects of environments and azimuth on lateral bias and lateral precision error. The mean value and standard deviation of lateral bias and lateral precision error were calculated and compared

Results

The results are shown in Figure 5. **A.** shows means errors in the localization task (localization bias). **B.** shows the statistical errors (lateral precision errors). Each curve represents a virtual environment.

Localization bias. We noticed some trends in the results. The localization of extreme positions (-90° ,

0° and 90°) were quite accurate regardless of environments. For instance, the localization bias for $+90^\circ$ was -8.03 degrees (± 9.56). For 0° , localization bias was between $+5.08^\circ$ and $+6.38^\circ$.

For intermediate positions, amplitude was higher. Indeed, subjects systematically overestimated the position. When the stimulus came from the left side, subjects tended to overestimate for the left. The same trend was observed for the right side.

Lateral precision error. The results showed intertrial constancy. Consequently, lateral precision error was quite stable between azimuths.

Effect of virtual environments. Bias localization and lateral precision error did not vary among the different environments. Statistical measurements confirmed those observations (localization bias: $F(2,10)=0.86$, $p = 0.45$; lateral precision error: $F(2,10) = 3.57$, $p=0.07$).

Discussion

The results were collected in a virtual reality environment, combining virtual sounds (from KU100 data base) and virtual environments (simulated with a HTC Vive helmet).

- Localization bias collected was higher for intermediate positions than extreme and central positions,

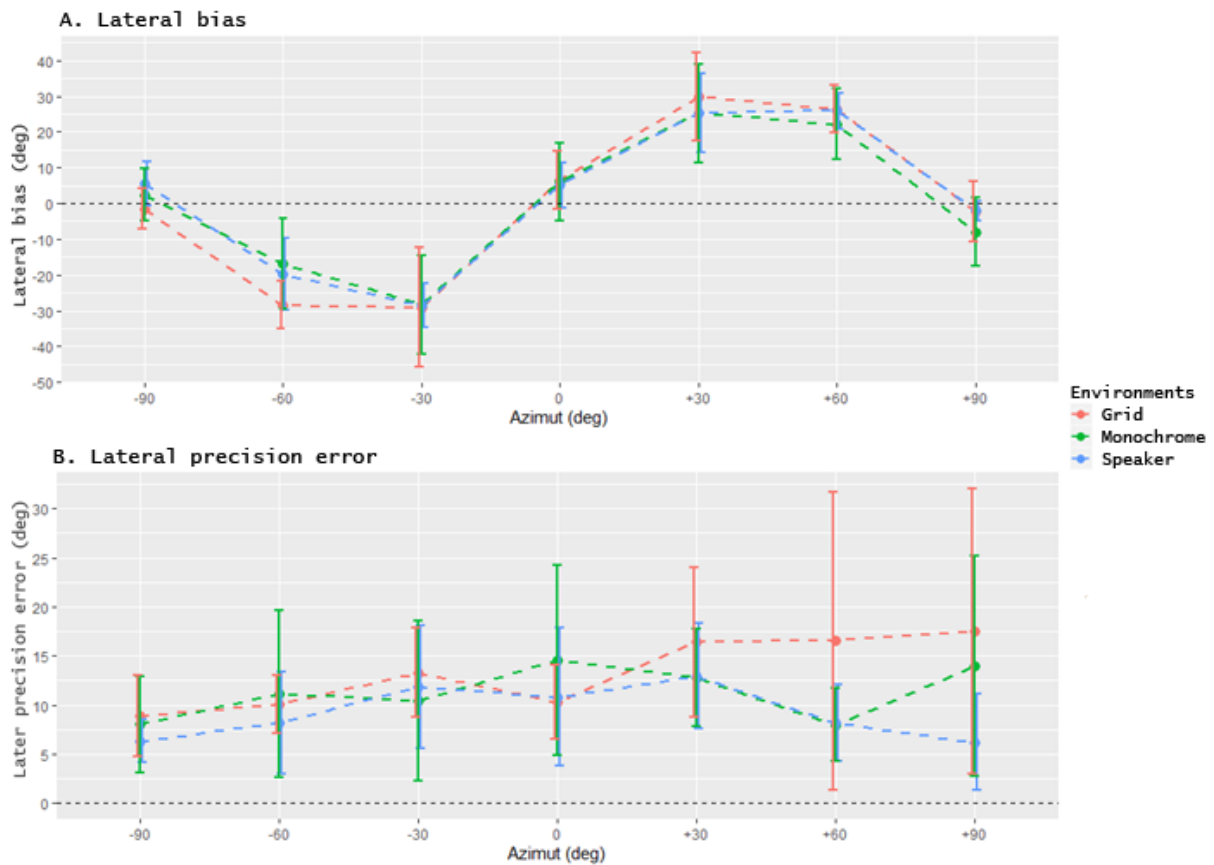


Figure 5 Results for experiment 2. Graphic A. reports the lateral bias. Graphic B. reports the lateral precision error. The error bars represent the standard errors.

- Lateral precision error was quite stable regardless of azimuth,
- Any differences were observed between virtual environments.

Effect of azimuth. Acoustic stimuli positions had an influence on localization bias. Previous studies show similar trends: higher estimation for intermediate positions than extreme and central positions (Makous & Middlebrooks (1990), Carlile et al., 1997; Lewald & Ehrenstein, 1998; Seeber, 2002; Oldfield & Parker, 1984).

Localization bias and lateral precision error amplitude. In the present study, localization biases were higher than in previous studies Makous & Middlebrooks (1990), Carlile et al., 1997; Lewald & Ehrenstein, 1998; Seeber, 2002; Oldfield & Parker, 1984).

Madjak et al. (2010) used individualized HRTFs in to evaluate localization bias and lateral precision error. Madjak et al.'s (2010) results are more accurate for localization bias, while lateral precision error is comparable to those of the present study.

Those observations may come from the data processing. Zero reference manipulation, for each subject, probably deteriorated localization bias. This

proposal was confirmed by lateral precision error results, which were quite similar to previous study. If that is the case, lateral precision error and localization bias measurements do not require individualized HRTFs.

Effect of virtual environment. Environments were created with more or fewer visual references. Bias localization and lateral precision error were not impacted by these visual references. Previous studies have shown that visual references improve localization abilities.

All the environments used in the present study included a 0° reference. This constant reference could explain the absence of differences among visual environments.

Conclusion

The initial hypothesis suggested that adding visual references would lead to an improvement in sound localization ability. Our results did not confirm this hypothesis. However, the results collected in experiment 2 are promising concerning the use of virtual generic sound in localization tasks.

GENERAL CONCLUSION

The results obtained in these 2 experiments were promising.

People suffering from hearing impairment frequently complain of difficulty in understanding speech, particularly in noisy situations. The relationship between poor hearing and the ability to distinguish sound sources are often mentioned.

Yet, the ability to localize sound in the hearing impaired are rarely tested. In the present study, sound localization ability of normal-hearing subjects was tested efficiently with the help of virtual environments.

Moreover, MAA results obtained in monaural conditions (in order to simulate unilateral hearing loss) matched the MAA of patients with unilateral hearing loss patients. This protocol could be a low-cost solution to evaluate MAAs clinically.

Experiment 2 did not show any influence of visual environment.

In the future, virtual environments could be more complex and more representative of everyday life (crowds, streets, noisy environment, *etc.*). Studies so far have mainly explored the effect of training in sound localization tasks, without visual references (Bauer et al., 1966; Kumpik et al., 2010) and with visual references (Recanzone, 1998; Lewald, 2002; Wozny and Shams, 2011).

In the short term, this system could be used as a support for training, in order to improve sound localization ability in hearing-impaired patients.

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